Project Synopsis: Next Word Prediction using LSTM & RNN

Next Word Prediction System using Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM)

1. Introduction

Next Word Prediction is a Natural Language Processing (NLP) task that predicts the most probable next word in a given sequence of text. This project leverages deep learning models, specifically **RNN and LSTM**, to understand language patterns and generate accurate predictions. Such systems are widely used in applications like autocomplete, chatbots, and virtual assistants.

2. Objective

- To build a model that predicts the next word in a sequence.
- To compare the performance of **RNN and LSTM models**.
- To understand sequential data processing using deep learning.
- To implement a real-time prediction interface using a simple application.

3. Dataset

- A custom text corpus is used.
- Text data is preprocessed by:
    - Converting to lowercase
    - Tokenization
    - Removing punctuation
- A **tokenizer (tokenizer.pkl)** is used to convert words into numerical sequences.

4. Methodology

Step 1: Data Preprocessing

- Tokenization using Keras Tokenizer
- Creating input sequences
- Padding sequences for uniform length

Step 2: Model Building

Two models are implemented:

1. RNN Model

- Simple Recurrent Neural Network
- Captures short-term dependencies

2. LSTM Model

- Advanced version of RNN
- Handles long-term dependencies using memory cells

Step 3: Training

- Models trained on sequences of text
- Loss function: Categorical Crossentropy
- Optimizer: Adam

Step 4: Model Saving

- rnn_model.h5 → RNN trained model
- lstm_model.h5 → LSTM trained model
- tokenizer.pkl → saved tokenizer

Step 5: Deployment

- app.py used to build an application (Streamlit)
- User inputs text → model predicts next word

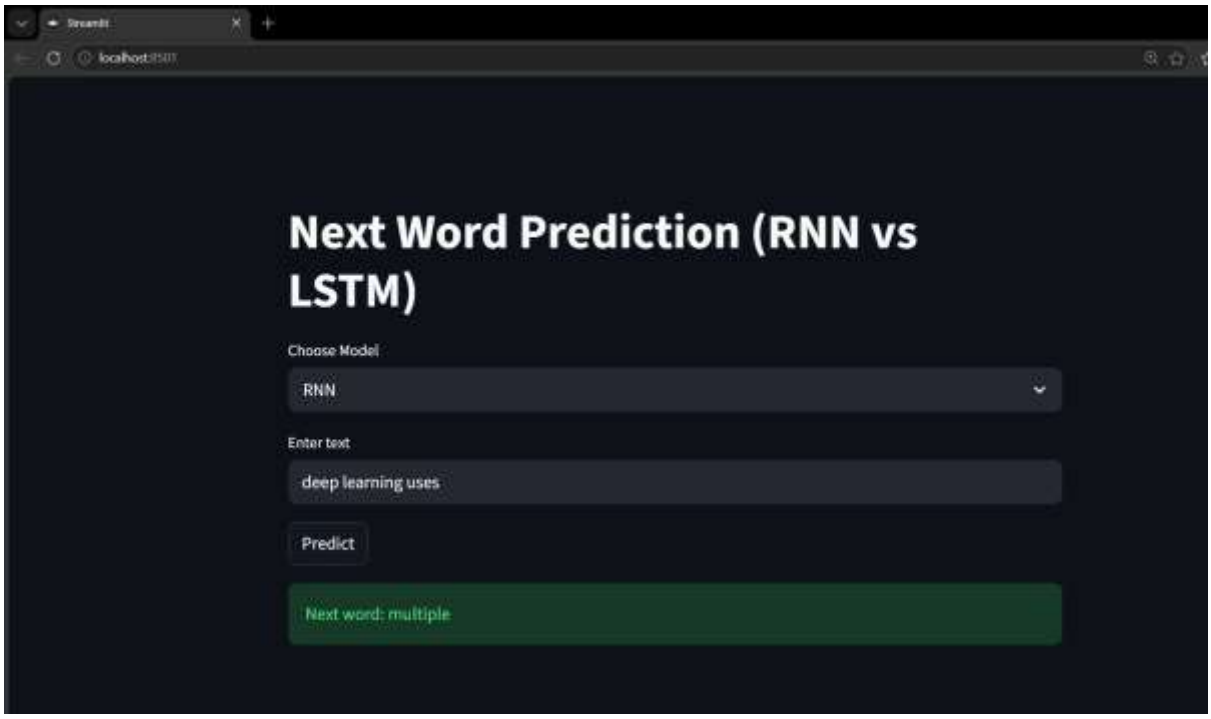
5. Tools & Technologies

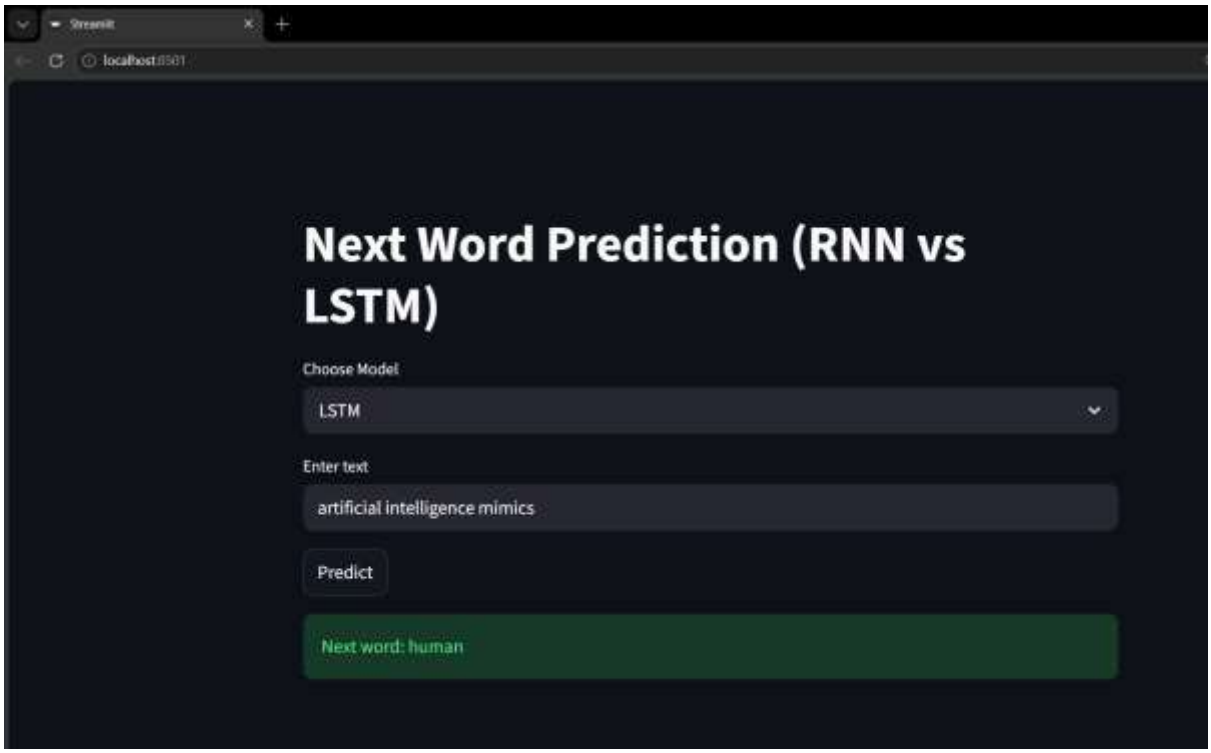
- **Programming Language:** Python
- **Libraries:** TensorFlow, Keras, NumPy, Pickle
- **Framework:** Streamlit
- **Environment:** Jupyter Notebook & VScode

7. Results

- LSTM performs better than RNN in most cases due to its ability to remember long-term dependencies.
- The model successfully predicts context-aware next words.
- The application provides real-time predictions.

OUTPUT:





8. Applications

- Text Autocomplete (like Google search)
- Chatbots & Virtual Assistants

- Email/SMS suggestion systems
- Content generation tools

9. Conclusion

This project demonstrates how deep learning models like RNN and LSTM can be effectively used for sequence prediction tasks in NLP. LSTM shows superior performance due to its ability to handle long-term dependencies, making it suitable for real-world language modeling applications.